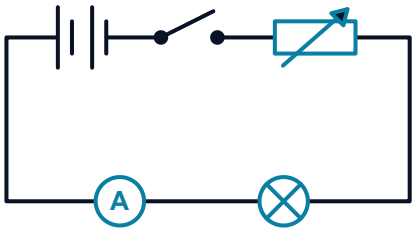
DO NOW · REQUIRED PRACTICAL

I–V characteristic of a filament lamp

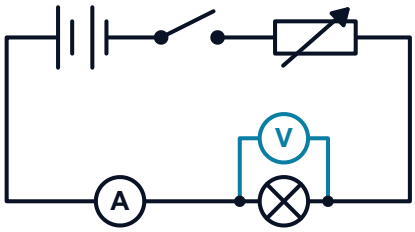
Electricity
Paper 1 · Combined and Separate
6 steps

A student finds the I–V characteristic of a filament lamp.

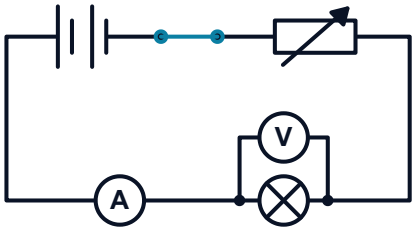
1 Connect the lamp in series with a power supply, a variable resistor and an ammeter.



2 Connect a voltmeter in parallel with the lamp.

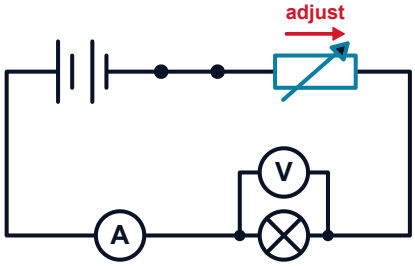


3 Record the current and the potential difference.



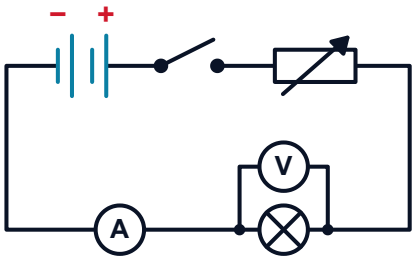
I = ___ A V = ___ V

4 Adjust the variable resistor and record a new pair of readings; do this several times.



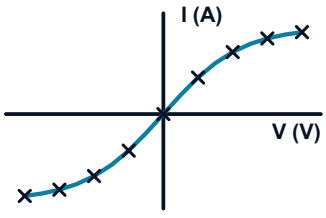
record a new pair of readings; repeat several times

5 Reverse the connections to the power supply and take the readings again.



reverse the connections to the supply

6 Plot a graph of current against potential difference.



I against V, both directions

© 2026 PanPhy Labs. All rights reserved.